MODULE 1: Generative & Probabilistic Models

🔹 Generative Models

Generative models are machine learning models that learn the underlying data distribution and generate new data similar to the training data. For example, they can generate realistic images, text, or audio. Instead of just classifying data, these models focus on creating new samples by learning probability distributions like $P(X)$or $P(X, Y)$.

🔹 Key Idea

The main idea behind generative models is to learn how data is distributed and then sample new data from that learned distribution. This allows the model to generate realistic outputs similar to real-world data.

🔹 Importance

Generative models are important because they help in data augmentation, enable creative AI applications like image and text generation, and support unsupervised learning where labeled data is not required.

🔹 Challenges

Generative models face challenges such as training instability, mode collapse where limited types of outputs are generated, high computational cost, and bias due to poor training data.

🔹 Probabilistic Models

✔ GMM

Gaussian Mixture Models assume that data is generated from a mixture of multiple Gaussian distributions. It uses the Expectation-Maximization (EM) algorithm and performs soft clustering, meaning each data point can belong to multiple clusters with different probabilities.

✔ HMM

Hidden Markov Models are used for sequence data where the system has hidden states and observable outputs. These models are widely used in applications like speech recognition and natural language processing.

✔ Bayesian Network

Bayesian Networks are probabilistic models represented using directed acyclic graphs, where nodes represent variables and edges represent dependencies. They use conditional probability to model relationships between variables.

✔ MRF

Markov Random Fields are undirected graphical models used to represent relationships between variables. They are commonly used in image processing and computer vision tasks.

✔ PGM

Probabilistic Graphical Models combine graph theory and probability theory to represent complex relationships and handle uncertainty in data efficiently.

MODULE 2: GAN

- **GAN**

Generative Adversarial Networks consist of two neural networks called the generator and the discriminator. The generator creates fake data samples, while the discriminator tries to distinguish between real and fake data.

- **Working**

GAN works based on a competition between the generator and discriminator, known as a min-max game. The generator tries to fool the discriminator, while the discriminator tries to correctly identify real and fake data.

- **Loss Function**

The GAN loss function is based on maximizing the probability of correct classification by the discriminator and minimizing the probability of detection by the generator.

- **Types of GAN**

Vanilla GAN is the basic version, while DCGAN uses convolutional layers for better image generation. WGAN improves training stability using Wasserstein distance. Conditional GAN generates data based on labels, and CycleGAN is used for image-to-image translation without paired data.

- **Problems**

GANs suffer from issues like mode collapse, where the generator produces limited outputs, and training instability due to the adversarial nature of learning.

- **Applications**

GANs are widely used in image generation, style transfer, deepfake creation, and data augmentation.

MODULE 3: VAE

- **VAE**

Variational Autoencoders are probabilistic models that generate data by learning latent representations. They are different from traditional autoencoders because they model data distribution.

- **Architecture**

VAE consists of an encoder that converts input into latent space and a decoder that reconstructs data from the latent representation.

- **Loss**

The loss function in VAE includes reconstruction loss and KL divergence, which ensures the latent space follows a normal distribution.

- **Latent Space**

The latent space in VAE is continuous and meaningful, allowing smooth generation of new data samples.

- **Types of Autoencoders**

Undercomplete autoencoders perform compression, sparse autoencoders activate only a few neurons, contractive autoencoders learn robust features, denoising autoencoders remove noise, and VAE is used for generative modeling.

MODULE 4: Transfer Learning

- **Transfer Learning**

Transfer learning is a technique where a pre-trained model is used for a new task. This reduces training time and improves performance when limited data is available.

- **Methods**

Feature extraction uses pre-trained layers without modification, while fine-tuning adjusts model weights for better performance on new data.

- **Advantages**

It requires less data, reduces training time, and improves model accuracy.

- **Advanced Topics**

Recent advancements include self-supervised learning, where models learn without labeled data, and meta-learning, where models learn how to learn.

MODULE 5: Ensemble Learning

🔹 Ensemble Learning

Ensemble learning combines multiple models to improve accuracy and performance compared to a single model.

🔹 Types

✔ Bagging

Bagging builds multiple models in parallel using different subsets of data, and Random Forest is a common example.

✔ Boosting

Boosting builds models sequentially, where each model focuses on correcting errors of previous models.

✔ AdaBoost

AdaBoost increases the weight of misclassified samples to improve performance.

✔ Stacking

Stacking combines predictions from multiple models using another model.

✔ XGBoost

XGBoost is an efficient gradient boosting algorithm that uses regularization to prevent overfitting and improve performance.

MODULE 6: Nascent Technologies in AI

- **2D Learning Limitations**

2D learning environments lack depth perception, provide passive learning experiences, and have limited interaction, making them less effective than immersive technologies.

- **AR & VR**

Augmented Reality overlays digital content onto the real world, while Virtual Reality creates a completely immersive virtual environment.

- **Metaverse**

The Metaverse is a virtual 3D world where users interact using avatars. It provides immersive and interactive experiences.

Characteristics

The key characteristics include persistence, immersion, real-time interaction, and interoperability.

- **Components**

The Metaverse consists of technologies like VR/AR, artificial intelligence, blockchain, and networking systems.

- **Challenges**

Major challenges include privacy concerns, lack of standardization, and high cost of implementation.

- **Opportunities**

It offers opportunities in education, training, simulation, and virtual collaboration.

- **AI + Quantum Computing**

Quantum computing represents the future of AI by enabling faster and more powerful computations. It uses concepts like qubits, superposition, and entanglement to solve complex problems more efficiently than classical computers.